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Application of a Jacobian-Free Newton-Krylov Method to the Simulation of Hypersonic Flows

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Many production-level hypersonic flow codes employ a defect-correction approach to solving the partial differential equations used to model high-speed thermochemical nonequilibrium flows. Although this approach typically provides fast convergence during the startup phase of a simulation, it is limited to a linear asymptotic convergence rate. Newton-Krylov methods are an alternative approach that can achieve a quadratic asymptotic convergence rate if an exact Jacobian is used in the implicit operator. The challenge is that the derivation of an exact linearization of a hypersonic flow code is non-trivial and prone to human error. In this paper, we present our work towards overcoming this challenge via the Jacobian-Free Newton-Krylov method. The method is Jacobian-free in the sense that the algorithm only requires matrix-vector products which are obtained by Fréchet derivatives. In our solver, we employ complex-step Fréchet derivatives. By doing so, the accuracy of our numerical derivatives are not limited by subtractive floating-point error, and as such we are able to get a Jacobian influence that is effectively as accurate as one that is analytically derived and numerically evaluated. We demonstrate the solver's performance on several test cases that are representative of the type of flows of interest to the hypersonics community. The convergence rate of the developed Newton-Krylov solver is compared against a point-implicit implementation based on a defect-correction method.

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I. Introduction

The analysis of thermochemical nonequilibrium flows using computational fluid dynamics (CFD) codes has become an important part of hypersonic vehicle design. For design work, typical flow modelling choices are either the Navier-Stokes (NS) or Reynolds-Averaged/Favre-Averaged Navier-Stokes (RANS) equations. RANS modelling provides a necessary trade-off between accuracy and efficiency that enables a CFD solver to be utilised in the aircraft design process [1]. The simulation of hypersonic, turbulent, reacting flows is challenging as a result of the complex physical models required to accurately capture the nonequilibrium effects [2]. In spite of these challenges, several production-level codes exist today that have successfully been used for the analysis of steady hypersonic flows [3–7]. These solvers share a common feature in that they are all based on the defect-correction approach [8], where the implicit operator is an approximate Jacobian constructed using a lower-order residual. The resulting system of linear equations is then iteratively solved using some form of implicit relaxation, such as Jacobi [9], Gauss-Seidel [10, 11], or Line Gauss-Seidel [12]. Although this approach typically provides fast convergence during the initial phase of a simulation, it ultimately suffers from a linear asymptotic convergence rate. This means that a large number of nonlinear steps is required to sufficiently converge the flow solution. The large computational cost is particularly problematic for design work employing numerical optimization, where many flow solutions are required to evaluate the performance objective of candidate designs [13], and large amounts of precision are required.

Newton's method is an alternative approach to the defect-correction method that can achieve a quadratic asymptotic convergence rate if an exact Jacobian is used in the implicit operator. However, deriving an exact linearization of a hypersonic flow code is difficult and prone to human error. Newton-Krylov methods are attractive for this application since they can be implemented in a Jacobian-free manner when employing a Krylov method such as the generalized minimum residual method (GMRES) [14]. This is due to a convenient property of GMRES: it only requires matrix-vector products when solving the linear system of equations. Jacobian-Free Newton-Krylov (JFNK) methods are a subset

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of Newton-Krylov methods that utilise Fréchet derivatives to approximate the matrix-vector products in the GMRES algorithm.

The JFNK method has been successfully applied in many branches of computational physics; Knoll et al. [15] provide an excellent survey of the field. In spite of this, it remains under-represented in hypersonic CFD codes. A potential reason for this could be the challenge associated with achieving sufficient accuracy of the matrix-vector products using Fréchet derivatives. For simulating thermochemical nonequilibrium flows, Olawsky et al. [16] observed that employing Fréchet derivatives was typically not stable and required some form of stabilization, such as using a first-order spatial discretization in the Jacobian. Rouzbar et al. [17] also used Fréchet derivatives for simulating chemical nonequilibrium flows; although no challenges were discussed, the results presented are observed to achieve only linear asymptotic convergence. Similar experiences have also been reported for low-speed flows [18]. Most recently Thompson et al. [19] and O'Connell et al.[20] have demonstrated the advantages of using a JFNK method utilising Fréchet derivatives for adaptive grid refinement, however the high-speed flows investigated were only modeled as a single-species, calorically perfect gas, that is, no high-temperature nonequilibrium effects were modelled.

In this paper, we present our work towards developing a JFNK method for simulating high-speed flows in thermochemical nonequilibrium. We have recently implemented a JFNK algorithm in our in-house flow solver, `Eilmer`. To overcome the challenge of achieving sufficiently accurate matrix-vector products with real-valued Fréchet derivatives, we employ a complex-step variant. This avoids accuracy degradation of the numerical derivatives due to subtraction operations with floating-point numbers, and as such we are able to get a Jacobian influence that is effectively as accurate as one that is analytically derived and evaluated. The main goal of this paper is to demonstrate the solver's performance on several test cases that are representative of the type of flows of interest to the hypersonics community: (1) blunt-body atmospheric entry flows; (2) flows with separated regions; and (3) supersonic combustor flows. The convergence rate of the developed Newton-Krylov solver is compared against a point-implicit implementation based on a defect-correction method. The formulation and implementation of the JFNK solver is described in Section II. Section III gives the performance comparisons on the three test cases mentioned above.

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II. Numerical Method

The `Eilmer` flow simulation software is a derivative of the Navier-Stokes solver described by Jacobs [21]. It has been further developed over the years, especially for high-temperature hypersonic flows, and with some significant effort toward verification by Gollan & Jacobs [22] and Damm [23]. The description of the `Eilmer` code in this section is kept brief with regards to the governing equations and discretization. A complete description is available in Gibbons et al. [24]. We do however devote some detailed discussion to the Jacobian-free Newton-Krylov implementation in Section II.B since that is the focus for this paper.

A. Governing equations and spatial discretization

`Eilmer` is formulated around the integral form of the conservation equations, which can be written as

$$\frac{\partial}{\partial t}\int_V \mathbf{U}\, dV = -\oint_S (\mathbf{F}_i - \mathbf{F}_v)\cdot \hat{n}\, dA + \int_V \mathbf{Q}\, dV \tag{1}$$

where S is the bounding surface and $\hat{n}$ is the outward-facing unit normal of the control surface. The symbol V in Eq.(1) is the volume of a cell while A is the area of the cell boundary. For a gas in thermal and chemical nonequilibrium governed by two temperatures, the vector of conserved quantities carried in a simulation is $\mathbf{U} = \left[\rho v_x, \rho v_y, \rho v_z, \rho E, \ldots, \rho_s, \ldots, \rho e_v\right]^T$ where the conserved quantities are respectively x-, y- and z-momentum per volume, total energy per volume, followed by n_{sp} individual densities for the chemical species and the mixture vibroelectronic energy per volume. For RANS simulations, there are extra transport equations and so extra quantities appear in $\mathbf{U}$ depending on the choice of turbulence model. See Gibbons et al. [24] for further details.

The finite volume method is used for spatial discretization of the governing equations. The conservation equations are applied to small polyhedra called cells. Flux values are estimated at midpoints of the cell interfaces and the integral conservation equation (1) is approximated as the algebraic expression

$$\frac{\partial \mathbf{U}}{\partial t} = -\frac{1}{V}\sum_{f}^{\text{faces}} (\mathbf{F}_i - \mathbf{F}_v)_f \cdot \hat{n}_f\, A_f + \mathbf{Q} \tag{2}$$

This form replaces the boundary integral with a sum over the faces, f, of the cell, and the fluxes with the equivalent averages over the face in question. Likewise the conserved quantities $\mathbf{U}$ and the source terms $\mathbf{Q}$ are the volume averages of the state inside the cell. The flow field thus consists of a collection of small cells, each containing values for the pressure, temperature, density, flow velocity, and any other quantity of interest within. These values are relaxed towards a steady state using an iteration process.

For steady state, we seek flow field values that satisfy $\partial\mathbf{U}/\partial t = 0$. Commonly, this is referred to as the residual, $\mathbf{R}$:

$$\mathbf{R}(\mathbf{U}) = \frac{\partial \mathbf{U}}{\partial t}. \tag{3}$$

The iterative relaxation algorithms are described in Sections II.B and II.C. Here we focus on the spatial estimations that go into computing the residual, $\mathbf{R}(\mathbf{U})$. The residual evaluation, in `Eilmer`, is broken into a number of substeps:

1) Reconstruct the flow on both sides of each face
2) Compute the inviscid fluxes $\mathbf{F}_i$
3) Compute the gradients of the flow at each face
4) Compute the viscous fluxes $\mathbf{F}_v$
5) Compute the source term $\mathbf{Q}$ in each cell
6) Assemble the residual $\mathbf{R} = \partial\mathbf{U}/\partial t$

Preceding flux calculation, flow field values are reconstructed either side of a cell face. In structured-grid mode, that reconstruction works along index-direction by index-direction. The scheme uses quadratic interpolants giving global second-order spatial accuracy when unlimited. The reconstructed values are limited using the van Albada limiter [25]. In unstructured-grid mode, an unweighted cell-centred least-squares gradient calculation is performed at each cell, and the reconstruction is performed via slope-limited extrapolation using the computed gradients. Refer to Gibbons et al. [24] for a fuller description of the reconstruction process in both structured and unstructured modes.

There is a variety of flux calculators implemented in `Eilmer` for computation of the inviscid fluxes. In this work, we have used the low-diffusion flux-splitting scheme by Edwards [26]. In particular, the implementation follows the description of the LDFSS method from Roy et al. [27], which is based on the LDFFS(2) variant from the original paper [26]. For flows with very strong gradients, such as blunt bodies in hypersonic flow, an adaptive calculator is used that selects the LDFSS(2) scheme away from shocks and the more dissipative LDFSS(0) scheme [26] near shocks. A shock detector is used to determine the switching between the two flux calculators. That shock detector is based on a simple measure of the relative change in normal velocity at interfaces. Specifically, we indicate a strong compression at the interface when

$$\frac{v_{R0} - v_{L0}}{\min(a_{R0}, a_{L0})} < \texttt{Tol} \ , \tag{4}$$

where `Tol` is the compression tolerance and is set at -0.5 in this work. This measure is applied to all interfaces in a block and then a second pass propagates the information to nearby interfaces. If a first cell interface is identified as having a strong compression, then LDFSS(0) is used for all interfaces attached to the cell containing that first interface.

Evaluation of the spatial derivatives of temperatures and velocities, needed for the viscous fluxes, is performed using a cell-centred least-squares gradient reconstruction. Details of that calculation are given in Section 2.3 of Gibbons et al. [24].

B. Globalized Newton-Krylov method

A Newton method is used for iterative relaxation of the flow field quantities. The goal is to satisfy $\mathbf{R}(\mathbf{U}) = 0$. A Newton linearization leads to nonlinear updates of the form:

$$\left(\frac{\partial \mathbf{R}}{\partial \mathbf{U}}\right)^n \Delta\mathbf{U}^n = -\mathbf{R}(\mathbf{U}^n), \quad \mathbf{U}^{n+1} = \mathbf{U}^n + \Delta\mathbf{U}^n, \qquad n = 0, 1, ... \tag{5}$$

Using the nonlinear updates directly as shown in Eq. 5 leads to some practical challenges in complex shock-dominated reacting flow systems. One challenge is robustness. We rarely start the iterations with such a good initial guess that we are in the quadratic region of convergence for Newton's method. A solution to this robustness issue is to employ pseudo-transient continuation with physicality checks. This is presented in Section II.B.1. Another challenge is efficiency. It is often inefficient to solve the linear system for the update, $\Delta\mathbf{U}^n$, in a completely accurate manner on each step of Newton's method. To do so can lead to "over-solving", especially if far from the region of convergence. We are only seeking an update to move the solution in the right direction; it does not need to be an exact update. As such, we

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can use an iterative linear solver and stop when a desired accuracy is reached. For our work, the Generalized Minimal Residual method (GMRES) is used as an iterative solver for the linear system, discussed in Section II.B.2. This is the Krylov part of the Newton-Krylov method for steady-state acceleration.

1. Pseudo-transient continuation

For robustness, as mentioned previously, a strategy is required to take an initial guess for the flow field into the region of convergence of Newton's method [15]. These strategies are known as globalization strategies. Standard globalisation strategies such as line search or trust-region methods often stagnate at local minima. This can be particularly problematic when the solution has complex features, such as shocks, not present in the initial condition [28]. Another strategy is pseudo-transient continuation. It is more favourable for these types of problems since it does not require a reduction in the global residual at each step [28]. In practical terms, Knoll et al. [15] describe it as being able to climb hills. Pseudo-transient continuation can be viewed as time-marching to the steady-state solution [29]. When choosing a time-marching method to globalize Newton's method, the implicit-Euler time-marching scheme is attractive: it is unconditionally stable; and its time linearization becomes Newton's method in the limit of infinite step sizes [30]. Deriving an implicit-Euler scheme begins with the semi-discrete Eq. 2 with a discrete residual:

$$\frac{\partial \mathbf{U}}{\partial t} = \mathbf{R}. \tag{6}$$

Using implicit-Euler time-integration, this can be written in fully discrete form as

$$\frac{\Delta \mathbf{U}^n}{\Delta t} = \mathbf{R}^{n+1}, \quad \text{where } \Delta \mathbf{U}^n = \mathbf{U}^{n+1} - \mathbf{U}^n, \tag{7}$$

and Δt is the discretized time increment. Since we do not know $\mathbf{R}^{n+1}$, Eq. 7 is linearised as

$$\frac{\Delta \mathbf{U}^n}{\Delta t} = \mathbf{R}^n + \frac{\partial \mathbf{R}^n}{\partial \mathbf{U}} \Delta \mathbf{U}^n. \tag{8}$$

This may then be rearranged to recover the implicit-Euler time marching iterate,

$$[\mathbf{A}]^n \Delta \mathbf{U}^n = \left\{ \frac{1}{\Delta t_{\text{local}}} \mathbf{I} - \frac{\partial \mathbf{R}^n}{\partial \mathbf{U}} \right\} \Delta \mathbf{U}^n = \mathbf{R}^n. \tag{9}$$

Note that as $1/\Delta t$ approaches 0, Newton's method (Eq. 5) is recovered. We introduced Δt_{local} above. This indicates the use of local time-stepping; we use a constant Courant-Friedrichs-Lewy (CFL) value across the domain and compute a corresponding Δt in each cell based on local cell conditions. For steady state, we are not interested in marching through time accurately, so we can use local time-stepping as part of the acceleration strategy. How the CFL value is updated is discussed in the paragraphs that follow. For efficiency and to avoid over-solving in the early parts of the updates, it is unnecessary to solve the linear system at each step exactly; we can solve inexactly. An inexact-Newtons method is applied [15, 30, 31], where an solution to the linear system in Eq. 9 is sought that satisfies

$$||\mathbf{R}^n - [\mathbf{A}]^n \Delta \mathbf{U}^n|| \leq \eta ||\mathbf{R}^n||, \tag{10}$$

were $\eta \in [0, 1)$ is called the forcing parameter. Although several methods are available for computing the forcing parameter [32], it is fairly common to use a constant forcing parameter [33] and we have chosen that approach.

After giving an initial CFL value, the CFL is adjusted automatically after each Newton step based on the change in residual:

$$CFL^n = CFL^{n-1} \left(\frac{||\mathbf{R}(\mathbf{U}^{n-1})||}{||\mathbf{R}(\mathbf{U}^n)||} \right)^a, \tag{11}$$

where a is some positive number, typically chosen as 1. This formulation allows the CFL to increase or reduce depending on convergence history: when the residual is dropping and things are going well, the CFL increases; and vice versa when the residual increases between steps.

To increase the robustness of the pseudo-transient continuation stepping, we apply a relaxation factor to the update. The relaxation factor is based on checking that flow quantities remain physically sensible and the relative change in mass in a cell does not exceed a prescribed amount, say 20%. This idea was presented as a damping scalar by Knoll and

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McHugh [34], and as a relaxation factor by Yildirim et al. [35]. We have adopted the nomenclature of the latter. Using this idea, the update at each Newton iterate is:

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \omega^n \Delta \mathbf{U}^n \tag{12}$$

where $\omega^n \in [0, 1]$. We loop over all cells to determine a minimum value of ω. That minimum value is based on finding a value of ω that does not allow the mass in any cell to change by more than a relative amount θ and ensuring that primitive flow values are physically realisable, i.e. density, pressure and temperature remain positive. This gives a calculation of ω^n as:

$$\omega^n = \min \left[1, \quad \frac{\theta U_{\text{mass}}}{\Delta U_{\text{mass}}}, \quad \omega_{\text{cell}} \quad \text{s.t.} \quad p \in p_{\text{phys}} \right] \tag{13}$$

When simulating multi-species flows, the relative mass change is applied on a per-species basis. Thus, the middle term in Eq. 13 is replaced with a check of over all species in each cell. We set a minimum allowable value for ω^n, typically 0.01. If ω^n equals or exceeds the minimum allowable, then we apply the update in Eq. 12. If ω^n is less than the minimum allowable, we reject that update, reduce the CFL and try again. Typically, the CFL is reduced to half its value after a rejected update step. We have found the use of this relaxation factor is critical in enhancing the usability of the solver on large flow fields with complex interacting flow features.

2. Krylov method

The Krylov term in the name Newton-Krylov comes from the use of a Krylov subspace linear solver for solving the linear system (Eq. 9) arising at each pseudo-transient Newton iteration. Several Krylov subspace methods exist [36, 37]. The Generalized Minimal RESidual (GMRES) algorithm [14] has been a favourable selection for CFD solvers [15, 38–41]. A primary factor in the popularity of GMRES is that the algorithm only requires a matrix-vector product, which may be approximated by a Fréchet derivative [42]. This suggests that the method can be matrix- or Jacobian-free. However, some approximation of the Jacobian is still required as a preconditioner to the linear system.

The linear system of equations presented as Eq. 9 are poorly scaled. For example, for turbulent flows, the turbulence variables can have residual components that are several orders of magnitude larger than the flow state variables. As a consequence of solving the linear system inexactly, the poor scaling can lead to sub-optimal convergence or even divergence since the system can be dominated by the largest residual component [33]. To improve the convergence characteristics of Eq. 9, residual vector (row) and solution vector (column) scaling is recommended [38]. The row and vector scaling outlined by Brown et al. [43] is applied to Eq. 9 in this work. The scaled system is given as

$$[\tilde{\mathbf{A}}]\Delta\tilde{\mathbf{U}} = \tilde{\mathbf{R}}, \quad \text{where} \quad [\tilde{\mathbf{A}}] = \mathbf{D}^{-1}[\mathbf{A}]\mathbf{D}, \quad \Delta\tilde{\mathbf{U}} = \mathbf{D}^{-1}\Delta\mathbf{U}, \quad \tilde{\mathbf{R}} = \mathbf{D}^{-1}\mathbf{R}. \tag{14}$$

The scaling matrix $\mathbf{D}$, which is a diagonal matrix, is made up of typical values for $\Delta\mathbf{U}$ [43, 44]. In our work, we choose characteristic scaling values as the absolute value of the maximum time rate of change in the various conservation equations. As such, we find a maximum time rate of change in species continuity, in the momenta equations, and in the energy conservation equations. The maxima across all momenta equations is used to scale values in $x-$, $y-$ and $z-$momentum equations. To avoid scaling by values close to zero, when the time rates of change are very small, we set a lower bound on the absolute value of the entries in $\mathbf{D}$ to be 1.

Explicit formation of the flow Jacobian, $\frac{\partial \mathbf{R}^n}{\partial \mathbf{U}}$, in Eq. 9 requires linearisation of the spatially second-order flow routines. This requires excessive storage and is computationally expensive to compute [45]. An advantage of GMRES over other linear solvers is that it requires only matrix-vector products, which may be approximated by a Fréchet derivative [42]. The Fréchet derivative is given as

$$\mathbf{J}\mathbf{v} \approx \frac{[\mathbf{R}(\boldsymbol{U} + h\mathbf{v}) - \mathbf{R}(\mathbf{U})]}{h}, \tag{15}$$

where $\mathbf{v}$ is a directional vector, h is the perturbation parameter, $\mathbf{R}$ is the residual vector, and $\mathbf{J}$ is the matrix operator, in this case the flow Jacobian. A difficulty of applying Eq. 15 directly is the choice of the perturbation parameter h. Considering truncation error, h should be chosen sufficiently small. However, if h is too small, then the round-off error from subtracting two large, almost equal numbers can degrade the accuracy of the approximation [46]. Erroneous matrix-vector products resulting from poor selection of h can lead to GMRES converging on the incorrect linear system solution at each step as discussed by Zingg et al. [47]. To relieve the difficulty of selecting an appropriate perturbation

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size, and to improve the accuracy of the evaluated derivatives, a complex-step approximation to the Fréchet derivative may be applied [48]. The complex-step approximation to the Fréchet derivative is written as

$$\mathbf{J}\mathbf{v} \approx \frac{\mathrm{Im}[\mathbf{R}(\mathbf{U} + h\mathbf{v}i)]}{h}. \tag{16}$$

In the complex-variable approach, no explicit subtraction is involved, and so the perturbation parameter may be chosen sufficiently small to solely control the truncation error. Our implementation in `Eilmer` allows for either real-valued or complex-valued evaluations of the Fréchet derivative. We prefer the complex-valued approach for its robustness and accuracy. There is, however, a memory cost in terms of storage of complex values as opposed to real values. For turbulent and reacting simulations, we find that the complex-valued Fréchet derivative is necessary to achieve suitable convergence.

The Krylov subspace methods tend to suffer from slow convergence for problems which arise from fluid dynamics [36]. In practice, methods such as GMRES require some form of preconditioning to achieve fast, robust convergence. Preconditioning transforms the system by either pre- or post-multiplying by a matrix which approximates $\mathbf{A}$. The former is referred to as left-preconditioning, and is applied as given

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{x} = \mathbf{P}^{-1}\mathbf{b}. \tag{17}$$

In left-preconditioning, the residual of the system is changed. This can be avoided by applying post-multiplying or right-preconditioning,

$$\mathbf{A}\mathbf{P}^{-1}\mathbf{P}\mathbf{x} = \mathbf{b}. \tag{18}$$

Right-preconditioning is used in this work. The precondition matrix $\mathbf{P}^{-1}$ should resemble $\mathbf{A}^{-1}$ as much as possible. $\mathbf{P}$ is typically chosen to be a reduced-order approximation of $\mathbf{A}$ which is then decomposed by some approximate factorisation. The factorisation of $\mathbf{P}$ enables fast inversion when performing the preconditioning. Selection of a suitable factorisation is challenging. In a review of factorisation methods for CFD applications, Saad et al. [49] found that no single method was superior to the rest. For hypersonic flows, a zero-fill Incomplete Lower-Upper (ILU) factorisation of an approximate first-order flow Jacobian has shown good results for inviscid, laminar and turbulent flows [50–52]. An ILU factorisation of a matrix is a sparse approximation of the LU factorisation; zero-fill denotes that the sparsity pattern of the ILU factorisation is that of the original matrix. Higher levels of fill are possible, typically denoted as ILU(p), where p is the level of fill. As the level of fill increases, the sparsity pattern becomes less like the original matrix and more like the exact factorisation. However, the added accuracy of the factorisation is balanced by the additional memory and computational expense. Several authors [38–41] have found higher fill levels, ranging from 2 through to 4, provided accelerated convergence for the simulation of subsonic airfoils. The effect of increasing the level of fill for a supersonic test case was briefly studied as part of this current work. In our testing, we found that the additional level of fill does not provide significant performance improvement over the ILU(0) preconditioner. As a consequence, for this work, the ILU(0) preconditioner is used.

The ILU preconditioner can be implemented in either a scalar or block flavour. In the block approach, the LU factorisation is applied per block, where the block structure is taken to be the natural block structure of the flow Jacobian. This is referred to as block-fill ILU or BFILU. Orkwis [53] found that BFILU was able to converge problems for which the scalar approach stalled. Pueyo [39] found that, by treating the zero entries within a block as if they were non-zeroes, the BFILU can be constructed via a scalar algorithm. This reduces the complexity of the code and is the approach applied in the current work. For domain decomposed problems, an additive-Schwarz [54] approach with no overlapping is employed.

The scaled right-preconditioned restarted GMRES algorithm implemented in `Eilmer` is presented in Figure 1. In this work, we have chosen to precondition first and scale second. Note that, in the calculation of $\tilde{J}\tilde{v}_j$, the Jacobian matrix $\left(J = \frac{\partial \mathbf{R}}{\partial \mathbf{U}}\right)$ is never explicitly formed. The product is calculated by first forming $z = \mathbf{D}^{-1}\tilde{v}_j$, then solving the linear system $\mathbf{P}w = z$, evaluating Jw via the complex Fréchet derivative, and then finally multiplying the result of the matrix-vector product by $\mathbf{D}$. The remainder of the GMRES algorithm resembles the standard method as outlined by Saad [14, 36].

C. Point-implicit relaxation: comparison method

To provide a comparison to the JFNK algorithm, we have tested against one of the simplest implicit methods: point-implicit relaxation. It is simple both in terms of mathematical formulation and implementation. In the point-implicit method, we solve Eq. 9. However, there are two key simplifications: (1) the Jacobian is formed per cell; and (2) the Jacobian is approximated with first-order spatial reconstruction. The update is applied cell-by-cell, with a direct matrix solve of the local linear system. We also use complex-step differentation when forming these local Jacobians.

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1. Start:

   - Choose x_0
   - Compute

     $r_0 = b - Ax_0$.
     $\tilde{r}_0 = Dr_0$.
     $\tilde{v}_1 = \dfrac{\tilde{r}_0}{||\tilde{r}_0||_2}$.

2. Iterate:

   - For $j = 1, ..., m$ do

     $J\tilde{v}_j = D(J(P^{-1}(D^{-1}\tilde{v}_j)))$.
     $\tilde{h}_{i,j} = (J\tilde{v}_j),\ \ i = 1, 2, ..., j$.
     $\tilde{v}_{j+1} = J\tilde{v}_j - \sum_{i=1}^{j} \tilde{h}_{i,j}\tilde{v}_i$.
     $\tilde{h}_{j+1,j} = ||\tilde{v}_{j+1}||_2$, and
     $\tilde{v}_{j+1} = \hat{\tilde{v}}_{j+1}/\tilde{h}_{j+1,j}$.

3. Solve the least-squares problem:

   - Define $\tilde{\tilde{H}}_m$ to be the $(m+1) \times m$ (Hessenberg) matrix whose nonzero entries are the coefficients $\tilde{h}_{ij}$, $1 \le i \le j+1$, $1 \le j \le m$ and $\tilde{V}_m \equiv [\tilde{v}_1, ..., \tilde{v}_m]$.
   - Find the vector $\tilde{y}_m$ which minimises $||\tilde{\beta}e_1 - \tilde{\tilde{H}}_m\tilde{y}||_2$ over all vectors $\tilde{y}$ in R^m.

4. Form approximate solution:

   - Compute $x_m = x_0 + P^{-1}D^{-1}\tilde{z}^m$, where $\tilde{z}^m = \tilde{V}_m\tilde{y}_m$.

5. Restart:

   - Compute $r_m = b - Ax_m$, if satisfied then stop,
   - else $x_0 \leftarrow x_m$ Go-to 1.

Fig. 1 A scaled right-preconditioned restarted GMRES algorithm.

III. Performance Comparisons

This section presents performance comparisons of the two relaxation algorithms (JFNK and point-implicit) on some representative simulation cases of interest in our work. The intent is to show relative performance towards steady-state convergence, so we have kept all other modelling and numeric selections the same. Also, because of the goal to show performance, we do not show grid converged results for each case. The grids used in these performance test cases are what would be considered 'medium' resolution grids in a typical sequence of coarse, medium and fine grids. The cases are selected to represent the variety of flows regularly simulated by `Eilmer` users. Those cases are: a blunt body in a two-temperature thermochemically reacting air stream; a slenderish body (double-cone) with separated flow in a two-temperature nitrogen stream; and turbulent reacting flow in a scramjet combustor.

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A. High enthalpy air flow over a cylinder

Experiments with high enthalpy flow of air over a cylinder were performed in the HEG by Karl et al. [55]. Simulations of these experiments are used as our first performance test case. These experiments were used in a CFD capability assessment for shock interactions by Knight et al. [56]. In the simulations here, we have followed the description in Ref. [56] and that work can be consulted for further details. In particular, we have concentrated on the 13.5 MJ/kg case, denoted as HEG condition III in Ref. [56] and labelled as run 627 in Reimann and Hannemann [57]. Table 1 summarises the simulation details for the performance test.

Table 1 Modelling and simulation details for the high-enthalpy flow of air over a cylinder.

	Modelling and simulation details	Citation
Physical modelling		
geometry	radius = 45 mm; span = 380 mm	
domain	2D geometric modelling; half cylinder (windward facing)	
gas model	two-temperature five-species air	
thermochemical model	Park's 2-T model	[58]
mass diffusion	Fick's first law with binary diffusion coefficients	
free stream conditions	$V = 4776\,\mathrm{m.s^{-1}}$; $T = T_v = 694\,\mathrm{K}$; $\rho = 3.26 \times 10^{-3}\,\mathrm{kg.m^{-3}}$; $Y_{N_2} = 0.7356$; $Y_{O_2} = 0.1340$; $Y_{NO} = 0.0509$; $Y_O = 0.0795$	Tab 1. [57]
initial condition	free-stream flow everywhere in the domain	
boundary conditions	supersonic inflow; extrapolation at outflow; fixed-temperature ($T_{\mathrm{wall}} = 300\,\mathrm{K}$) no-slip non-catalytic wall	
Grid and blocking		
type	structured grid [see Fig. 2(a)]	
topology consideration	grid aligned with bow shock to enable accurate heat flux estimation	
cell count	9520 quads (80 circumferential × 119 radial)	
height of first cell off wall	1.0 μm	
blocking	10 blocks partitioned using METIS	[59]
Spatial numerics		
mode	unstructured	
flux calculator	adaptive LDFSS(0)/LDFSS(2) flux calculator	[26]
limiter	Venkat augmented with Park's heuristic pressure-based limiter	[60], [61]
JFNK settings		
initial CFL	1.0	
auto CFL start	immediately from first step	
target for linear solve, η	0.1	

Figure 2 shows the numerical simulation domain and steady-state flow field results obtained with the JFNK algorithm. The half cylinder domain and structured grid with 9520 cells is shown in Fig. 2(a). In Fig. 2(b) two maps of flow field quantities are shown at steady state: the vibrational temperature (in top half) and the mass fraction of monatomic nitrogen (in bottom half). Note the flow field is essentially symmetric about an axis aligned with the free-stream direction. However, we have chosen to simulate the full windward portion of the cylinder because this alleviates a numerical difficulty when estimating heat transfer at the symmetry plane. Comparisons of pressure and heat transfer distributions between simulation (`Eilmer`) and the experiment are shown in Fig. 3. We consider this good agreement.

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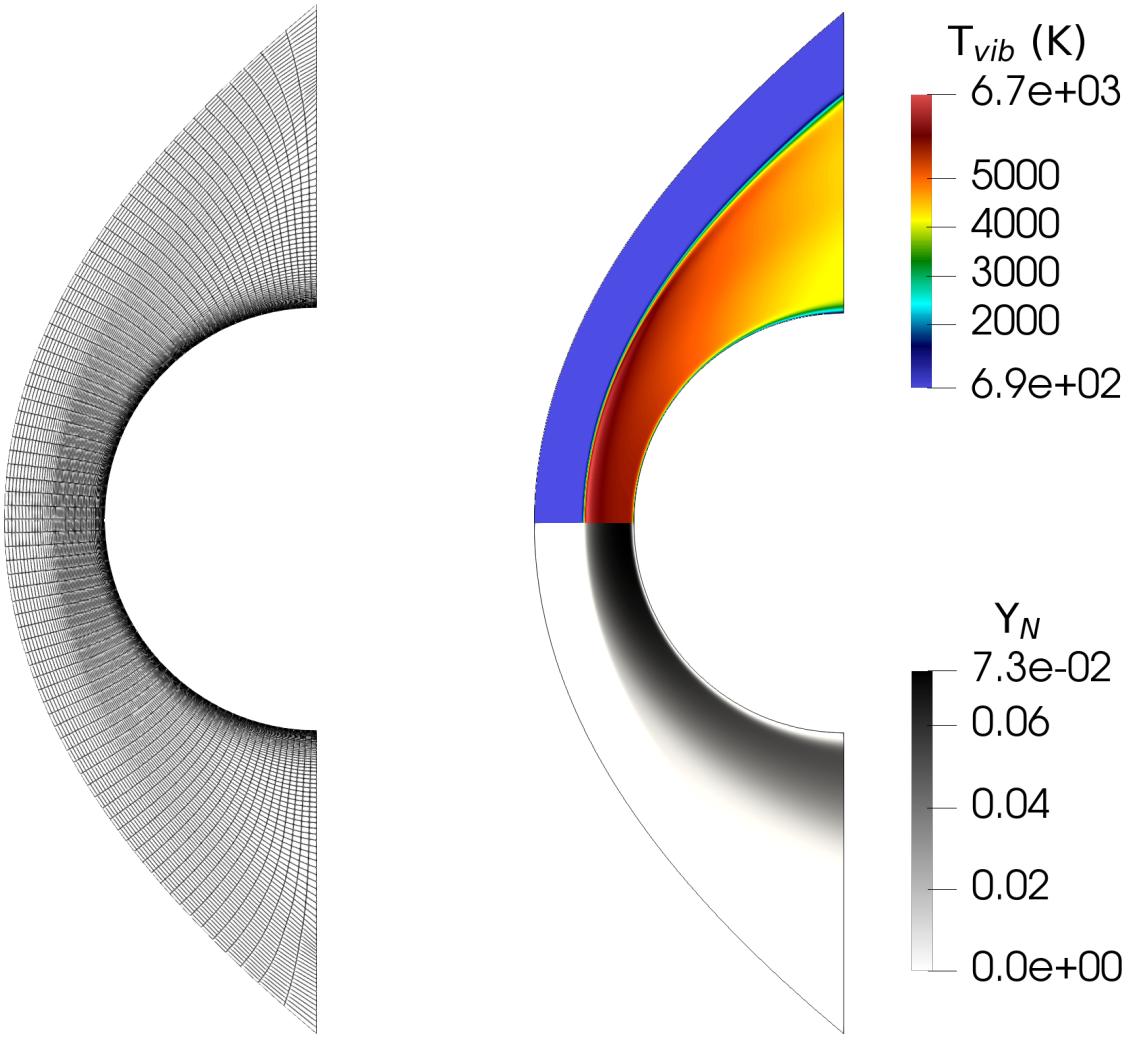
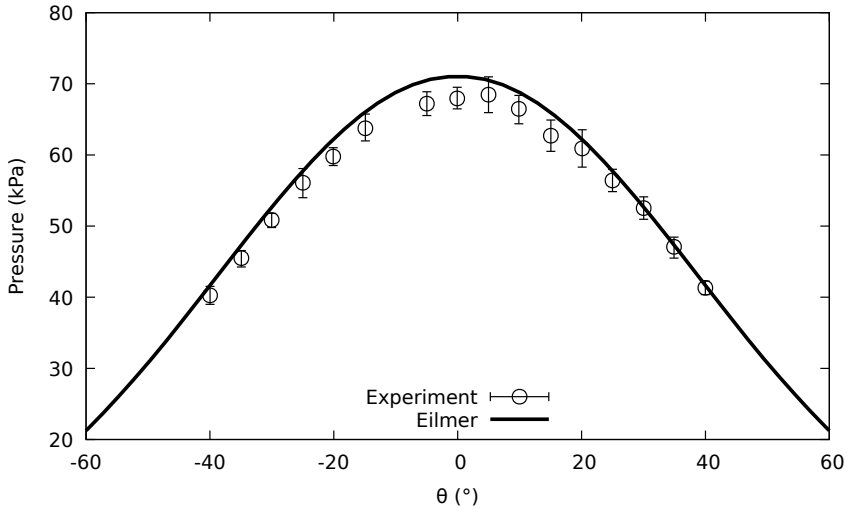
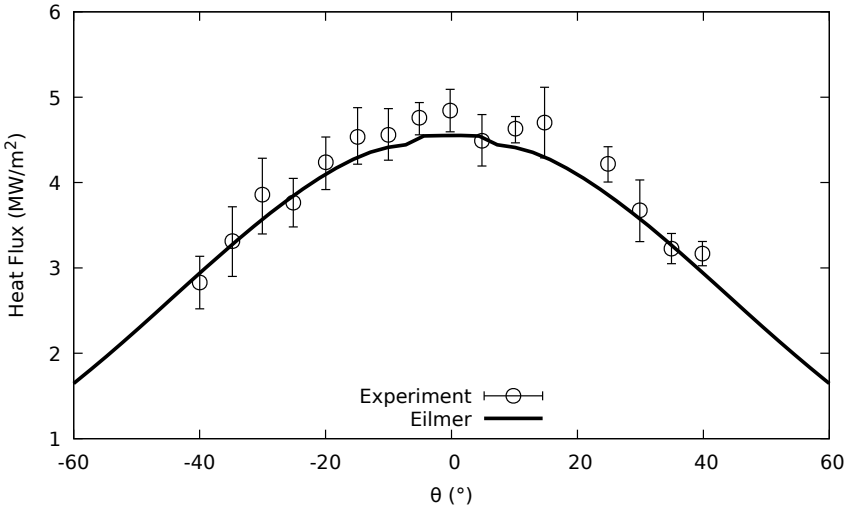


Fig. 2 Visualisation of simulation of high enthalpy air flow over a cylinder: (a) half cylinder domain with structured grid; and (b) *TOP* – vibrational temperature map; *BOTTOM* – mass fraction of monatomic nitrogen, *N*

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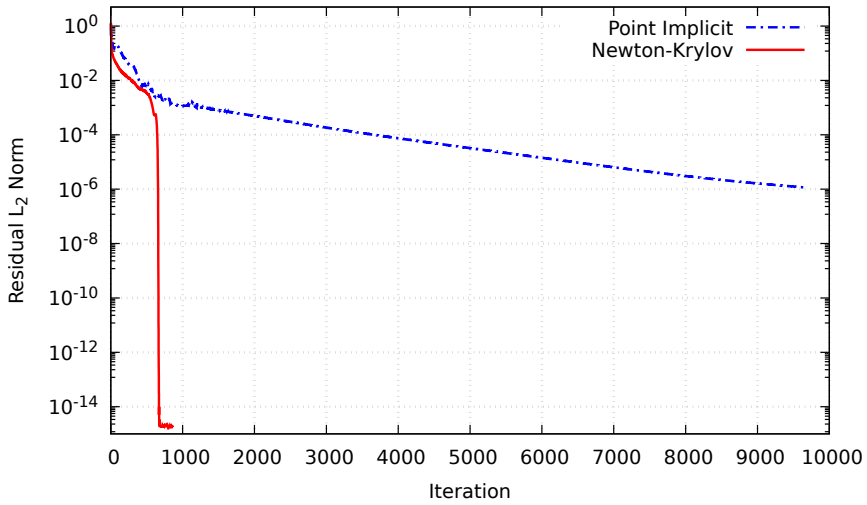
(a) pressure on cylinder surface



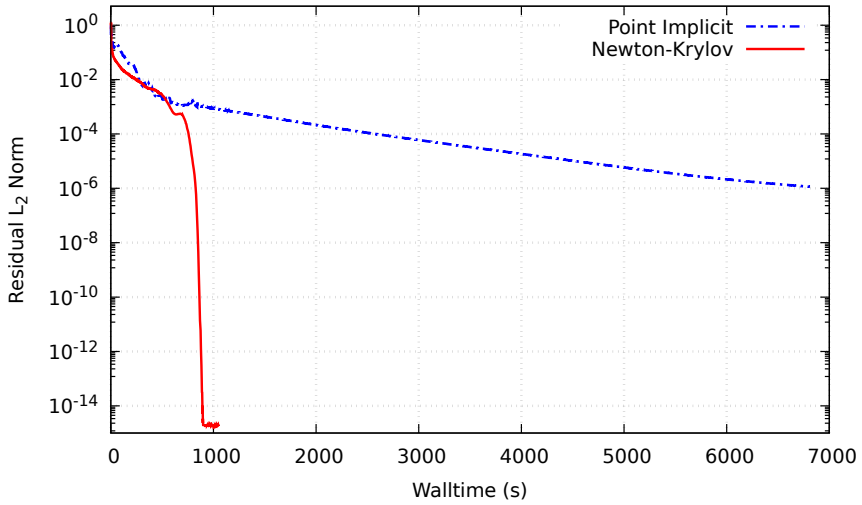
(b) heat transfer on cylinder surface

Fig. 3 Numerical simulation and experiment compared: (a) pressure on surface; (b) heat transfer on surface.

The interesting result, related to the intent of this paper, is the performance comparison shown in Fig. 4. Figure 4(a) compares the relative residual drop of the L_2 norm of all components of the residual as a function of iteration count. The JFNK method converges more the 14 orders of magnitude drop (relative to an initial residual) before reaching numerical buzz at machine precision. The quadratic convergence is evident from around the 500th interation onwards. By contrast, the point-implicit method was stopped early when it reached a drop of 6 orders of magnitude at around 9500 iterations. Each JFNK iteration is more costly than a point-implicit update, so we also examined performance in terms of wall-clock time. This is shown in Fig. 4(b). Even on this metric, the JFNK algorithm has superior run-time performance for achieving a steady state as compared to the point-implicit algorithm. The JFNK algorithm achieves machine convergence in less than 1000 seconds on 10 cores of a AMD Ryzen Threadripper Pro 3975. (That CPU is used in all the simulation cases in this work.)



(a) residual drop with iterations



(b) residual drop with wall-time

Fig. 4 Performance comparison between JFNK algorithm and point-implicit algorithm for high-enthalpy flow of air over a cylinder.

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B. Nitrogen flow over a double cone

The second case used for performance testing is the flow of nitrogen over a double cone. The particular geometry and flow conditions are Run 35 taken from experiments performed at CUBRC [62]. We have used an updated set of nonequilibrium free stream conditions reported by Candler et al. [63], which were the result of CFD simulations of the facility nozzle to determine the extent of vibrational nonequilibrium. As a performance test, we are interested in this flow because it exhibits a separated flow region between the first and second cone. This separated flow can cause issues with convergence so we consider this a good test for the JFNK algorithm. The full details of the simulation are presented in Table 2.

Table 2 Modelling and simulation details for nitrogen flow over a double cone.

Modelling and simulation details		Citation
Physical modelling		
geometry	$\theta_1 = 25°$; $\theta_2 = 55°$	[62]
domain	axisymmetric	
gas model	two-temperature two-species nitrogen	
thermochemical model	nitrogen rates from Park's 2-T model	[58]
free stream conditions	$V = 2571\,\mathrm{m.s^{-1}}$; $T = 99.38\,\mathrm{K}$; $T_v = 2768\,\mathrm{K}$; $\rho = 6.12 \times 10^{-4}\,\mathrm{kg.m^{-3}}$; $Y_{N_2} = 1.0$	Tab 3. [63]
initial condition	free-stream flow everywhere in the domain	
boundary conditions	supersonic inflow; extrapolation at outflow; fixed-temperature ($T_{\mathrm{wall}} = 300\,\mathrm{K}$) no-slip non-catalytic wall	
Grid and blocking		
type	structured grid [see Fig. 5(a)]	
cell count	46,746 quads (264 along body × 159 normal to body)	
height of first cell off wall	$1.0\,\mu$m	
blocking	32 blocks partitioned using METIS	[59]
Spatial numerics		
mode	unstructured	
flux calculator	LDFSS(2) flux calculator	[26]
limiter	Venkat augmented with Park's heuristic pressure-based limiter	[60], [61]
JFNK settings		
initial CFL	0.1	
auto CFL start	constant CFL until global relative residual had dropped to 0.5	
target for linear solve, η	0.1	

A depiction of the double cone simulation is given in Fig. 5. Figure 5(a) shows the axisymmetric domain and structured grid. The JFNK-converged steady-state flow field is shown in Fig. 5(b). The Mach number field is plotted, and overlayed are contours of vibrational temperature. The separated region, between the first and second cone, is evident in the contours of vibrational temperature.

There is generally good agreement between the simulation and experiment, as shown in Fig. 6. The extent of the separated region is smaller in the simulation as compared to the experiment. This is evident in both pressure and heat transfer measurements. This agreement might be improved with further grid refinement or more sophisticated modelling, such as partial slip wall effects on the first cone [63]. However, in this paper, we do not go any further to investigate this difference in separated region length since we are focussed on the performance comparison.

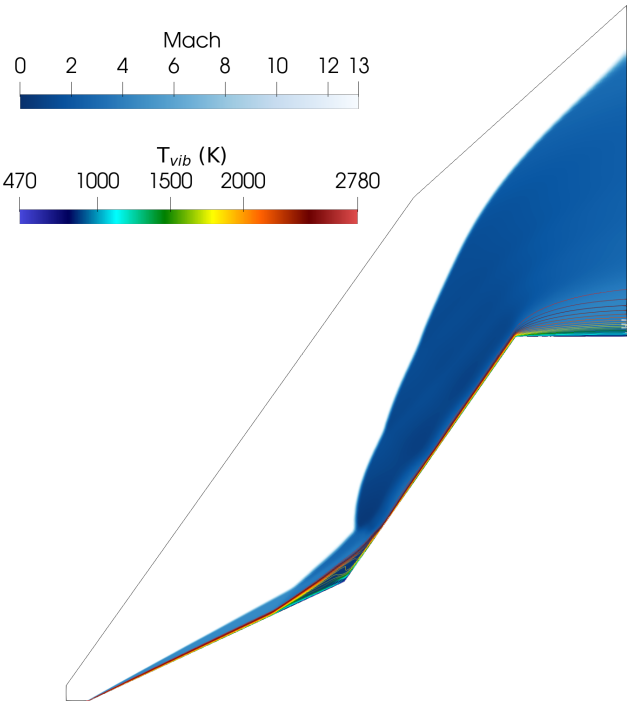
The performance comparison between the JFNK algorithm and the point-implicit algorithm is quite dramatic for this case: JFNK converges in 1050 iterations; the point-implicit fails to converge. This result is shown in both in terms of iterations and wall-clock time in Fig. 7. Comparing the JFNK convergence history for the double cone and the cylinder, there is a noisier path to deep convergence for the double cone. Possible reasons for this noisy convergence are difficulties establishing flow structures in the sensitive separated flow region and/or limiter ringing.

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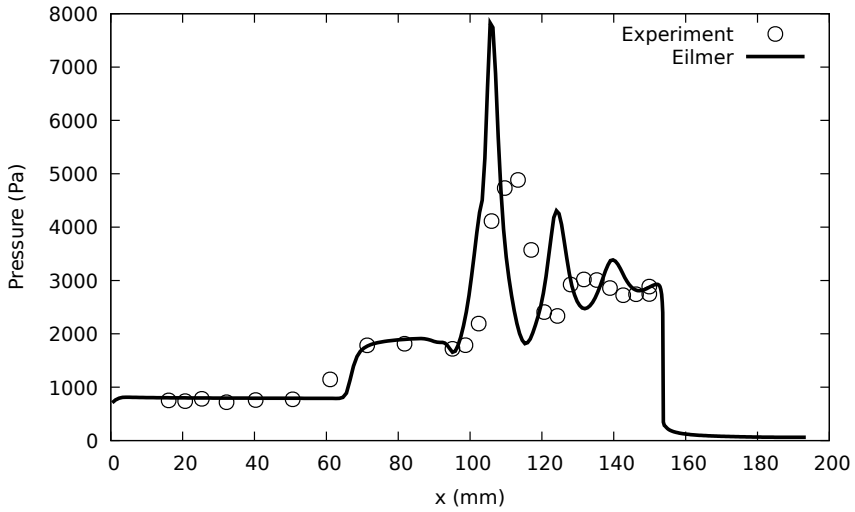
(a)



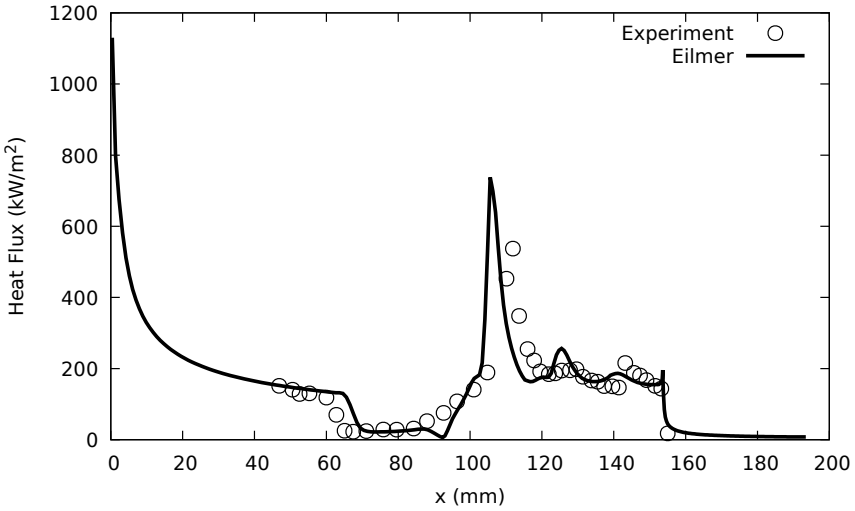
(b)

Fig. 5 Visualisation of simulation of nitrogen flow over a double cone: (a) axisymmetric domain with structured grid; and (b) Mach number heat map with contours of vibrational temperature overlayed

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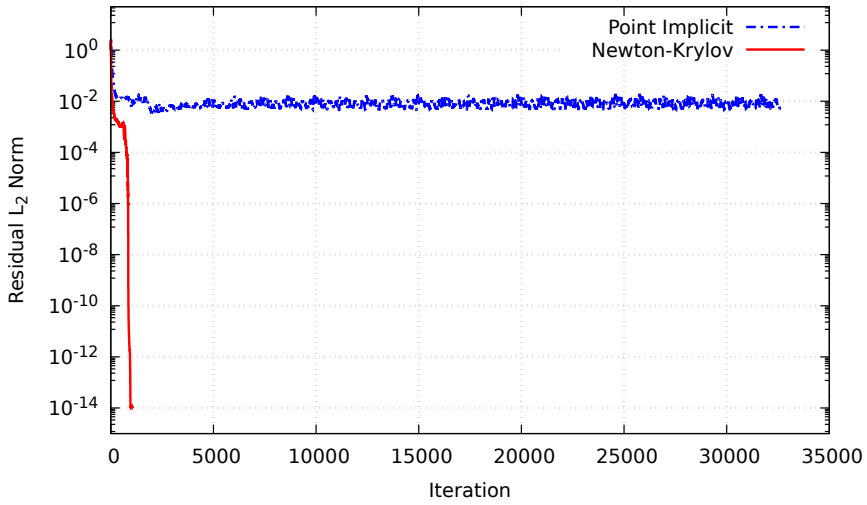
(a) pressure on double cone surface



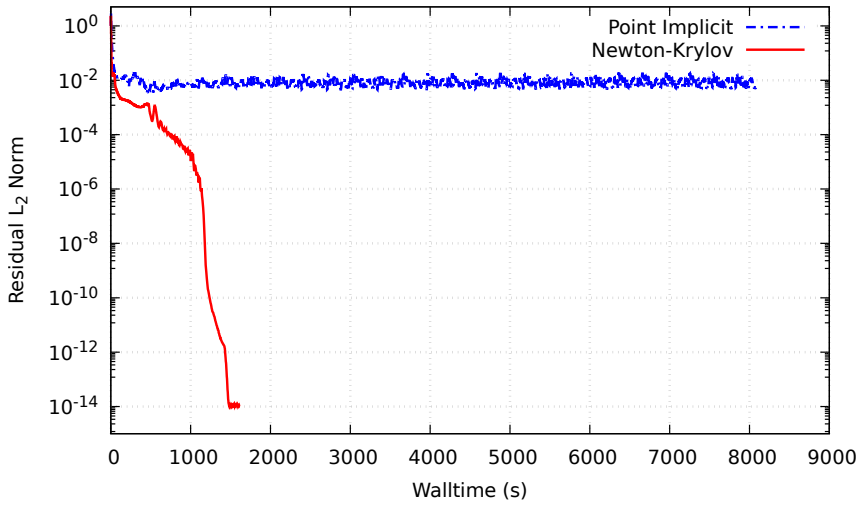
(b) heat transfer on double cone surface

Fig. 6 Numerical simulation and experiment compared for nitrogen flow over a double cone: (a) pressure on surface; (b) heat transfer on surface.

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(a) residual drop with iterations



(b) residual drop with wall-time

Fig. 7 Performance comparison between JFNK algorithm and point-implicit algorithm for nitrogen flow over a double cone.

C. Reacting flow in a supersonic combustor

The Burrows-Kurkov supersonic combustor experiments [64] are frequently used for validation in the high-speed reacting flow CFD community. They are the subject for this third performance comparison test. These experiments have been simulated using an earlier version of `Eilmer` by Hoste et al. [65], and that work gives a complete description of the numerical setup. We have attempted to replicate that setup. However, in this work, we have two differences from that of Hoste: the use of a hybrid structured-unstructured grid (see Fig. 8), and no explicit model for turbulence-chemistry interaction. We are not concerned by the lack of turbulence-chemistry interaction model since our focus is on how well (or otherwise) the JFNK algorithm performs in a high-speed flow with exothermic reactions. The answer to that question is very well. A summary of the modelling selections is given in Table 3 and the flow field at converged steady state is shown in Fig. 9.

Table 3 Modelling and simulation details for reacting flow in a supersonic combustor.

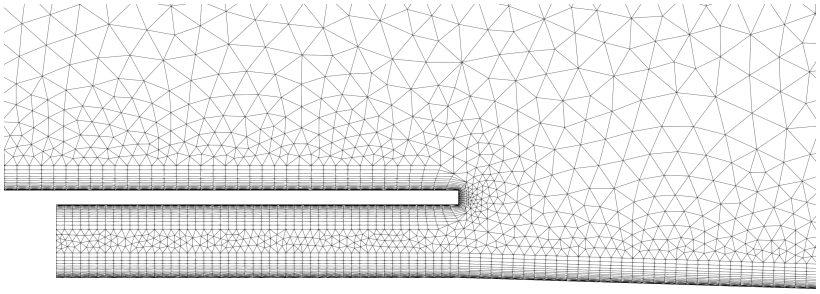
	Modelling and simulation details	Citation
Physical modelling		
geometry	combustor with step, injection behind step tangential to flow	[64]
domain	2D	
gas model	thermally perfect air + hydrogen combustion species	
chemistry model	7-species, 8-reactions air-H_2 combustion	[66]
mass diffusion	Fick's first law; constant Lewis number for diffusion coefficients	
turbulence model	Spalart-Allmaras	[67]
turbulent parameters	$\mathrm{Sc}_{\mathrm{turb}} = 0.7$; $\mathrm{Pr}_{\mathrm{turb}} = 0.85$	
free stream conditions	$V = 1741\,\mathrm{m.s^{-1}}$; $T = 1238\,\mathrm{K}$; $p = 96\,\mathrm{kPa}$; $Y_{N_2} = 0.486$; $Y_{O_2} = 0.258$; $Y_{H_2O} = 0.256$	[64]
injectant gas conditions	$V = 1217\,\mathrm{m.s^{-1}}$; $T = 254\,\mathrm{K}$; $p = 101.4\,\mathrm{kPa}$; $Y_{H_2} = 1.0$	[64]
initial condition	free-stream flow everywhere in the domain	
boundary conditions	supersonic inflow; extrapolation at outflow; fixed-temperature ($T_{\mathrm{wall}} = 300\,\mathrm{K}$) no-slip non-catalytic wall	
Grid and blocking		
type	unstructured grid [see Fig. 8]	
cell count	46,221 quads and 85,238 tris	
grid generation	Pointwise's T-REX to grow structured boundary layer grid into unstructured away from wall	
height of first cell off wall	$1.0\,\mu\mathrm{m}$	
blocking	32 blocks partitioned using METIS	[59]
Spatial numerics		
mode	unstructured	
flux calculator	LDFSS(2) flux calculator	[26],[27]
limiter	Venkat MLP limiter	[60], [68]
JFNK settings		
initial CFL	0.5	
auto CFL start	immediately from first step	
target for linear solve, η	0.01	

The performance comparisons for this supersonic combustor case is shown in Fig. 10. Once again, the result is particularly favorable for the JFNK algorithm. The JFNK algorithm reduces the relative residual to machine precision in fewer than 500 iterations. The point-implicit had not stalled when we terminated the simulation, but it was only making slow progress towards convergence. For the point-implicit, the CFL was held at 5.0. Attempts to push this value larger led to numerical instability. Returning to the JFNK convergence, the complexities of this flow field cause the algorithm to require a larger number of iterations, compared to the earlier cases, to push into the region of quadratic convergence.

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(a) whole of domain with grid



(b) zoom in near step injector

Fig. 8 Domain and grid for supersonic combustor simulation.

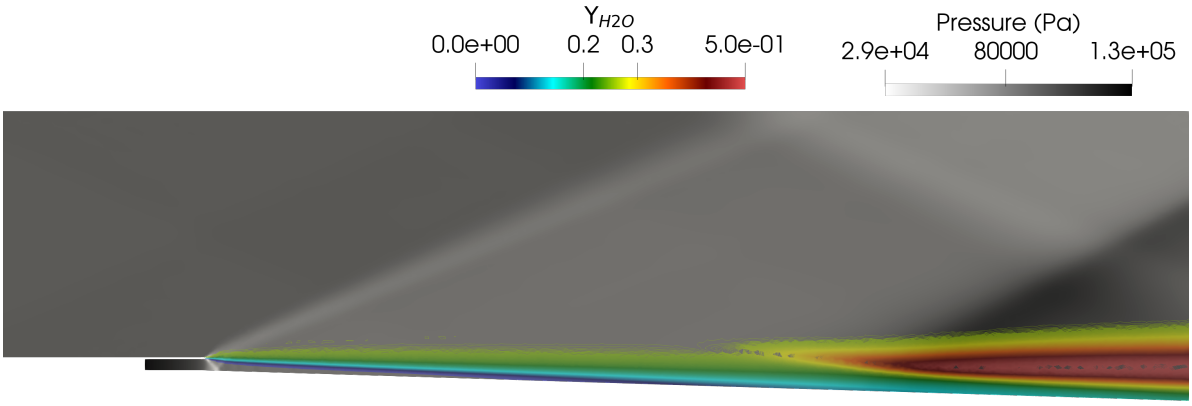
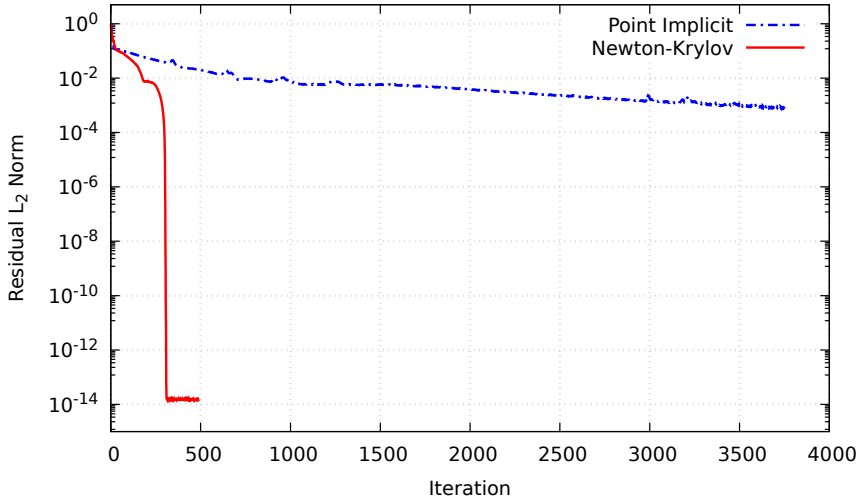


Fig. 9 Steady state flow field for supersonic combustor. Colors show water (H_2O) mass fraction. Grayscale is pressure contours.

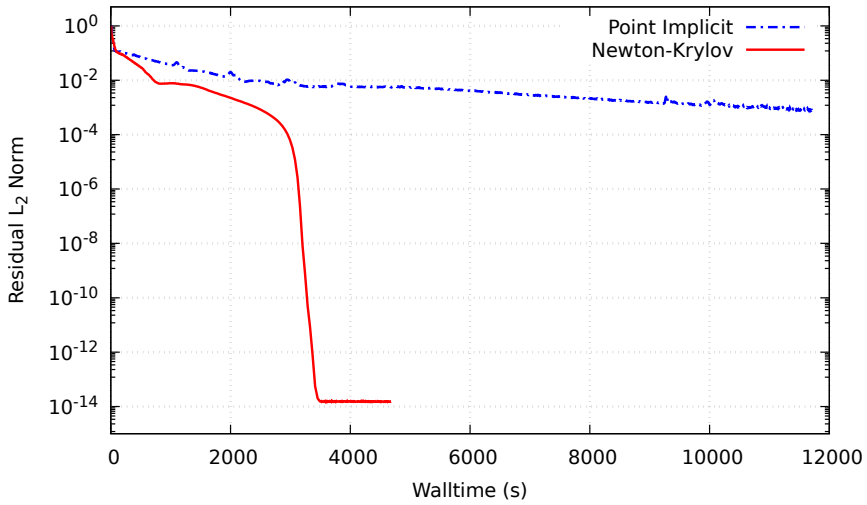
Here the complexities are establishment of the injector jet, mixing layer and associated chemical reactions.

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(a) residual drop with iterations



(b) residual drop with wall-time

Fig. 10 Performance comparison between JFNK algorithm and point-implicit algorithm for a supersonic combustion reactor.

IV. Concluding Remarks

In this work, we have presented a Jacobian-free Newton-Krylov (JFNK) algorithm applied in hypersonic flows. The distinctive nature of hypersonic flows means that the algorithm needs to work across a variety of thermochemical models: ideal gases, thermally perfect gases with finite-rate chemistry, and multi-temperature gases. We needed to avoid complicating how the Jacobian is formed in the presence of this variety of gas models. We found that GMRES and Fréchet derivatives via complex-valued differentiation were key enablers: with GMRES, we never explicitly form the Jacobian, only Jacobian-vector products; and complex-valued Fréchet derivatives give an accurate and robust representation of the Jacobian regardless of flux calculator, gas model, turbulence model, etc. We have not seen this use of complex numbers in a Newton-Krylov method widely applied in hypersonic flow solvers.

The performance of the JFNK algorithm was compared to a simple, yet standard algorithm: point-implicit relaxation. We chose three simulation cases to emphasise the point about robustness across a variety of gas models, and because they are of regular interest to users of `Eilmer`. Those three simulation cases were: high enthalpy air flow over a cylinder; two-temperature nitrogen flow over a double cone (with separation); and turbulent reacting flow in a supersonic combustor. The JFNK algorithm performed admirably in all cases, achieving machine precision with at least 14 orders of magnitude relative residual drop. By comparison, the point-implicit was slow to converge on two cases and completely stalled on a third. We compared both iteration count and wall-clock time; the JFNK outperformed the point-implicit on both measures for every case.

All of the simulation cases presented here were two-dimensional. We have preliminary results that indicate that the JFNK algorithm works very well in three dimensions also [23, 24]. However, at very large scales, we may need to consider degradation due to block decomposition and how this might be mitigated with more sophisticated preconditioners. This will be the subject of future work.

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Acknowledgments

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